FORMAT HUMAN OPTIMIZED

Part XI: Pepsi Black — CO_2 SVEs Mimicking Dolphins in the Black Sea

Video Experiment 14: youtu.be/FBsfnDhfJIk

14. Pepsi Black — "Gas Bubbles" That Jump, Dive, and Sink Like Dolphins

What was tested:

Chilled Pepsi Black was poured into a standard glass cup. No heating, no manipulation—just a carbonated beverage at ambient conditions. Recorded in slow motion on an iPhone. The question was simple: what do the "CO_2 bubbles" actually do when you watch them closely enough?

What standard thermodynamics predicts:

CO_2 dissolved under pressure comes out of solution as gas bubbles when the container is opened. These bubbles nucleate on surface imperfections, grow, detach, rise through the liquid (buoyancy: gas density $\ll$ liquid density), reach the liquid-air interface, and pop—the thin liquid film surrounding the gas stretches beyond its mechanical limit and ruptures. Surface tension is the reason the bubble exists; breaching the surface is how it dies. The gas disperses into the atmosphere. End of story. A popped bubble does not reassemble, does not arc through the air, does not dive back into the liquid, and certainly does not sink.

What was observed:

Bubbles that don't pop. Entities reach the liquid-air interface and cross it—breaching the surface tension barrier—without rupturing. They maintain their spherical integrity on the other side. This is the structural equivalent of a submarine surfacing with its hull intact. The very mechanism that should destroy them (surface tension breach) leaves them untouched.

Dolphins in the Black Sea. A single SVE was tracked across 40 consecutive frames. It ascended through the liquid, breached the surface intact, executed a complete ballistic arc through the air—rising to a remarkable height above the liquid—reached its apex, descended back toward the surface, and *dove back into the liquid.* Re-entry. Like a dolphin jumping out of the ocean and diving back in. The entity maintained structural integrity through the entire cycle: liquid $\rightarrow$ air $\rightarrow$ liquid.

They sink. After re-entry, the SVE did not float back to the surface. It sank. Progressively deeper, frame after frame, until it disappeared into the dark liquid. A CO_2 gas bubble in Pepsi has a density roughly 577 times lower than the surrounding liquid. Buoyancy would launch it back to the surface like a cork. Instead, it sinks. Because it is not a gas bubble.

A whole population doing this. The tracked SVE is not special. The full video shows dozens of entities walking across the surface, bouncing like stones skipped across a lake, jumping, and diving. This is not an isolated artifact—it is what "carbonation" actually looks like when you slow down the camera.

What this means:

The "fizz" in your soda is not gas. The entities emerging from carbonated beverages behave as solid or semi-solid structures—they have mechanical cohesion, they follow ballistic trajectories, they survive repeated interface crossings, and they sink in a liquid 577 times denser than the gas

they supposedly are. Every one of these behaviors is impossible for a free gas pocket.

This is the most accessible experiment in the Kitchen Thermodynamics series. No stove, no thermometer, no special equipment. A bottle of Pepsi Black, a glass, and a phone in slow-motion mode. The anomaly is the fizz itself. It has been in front of every human being who has ever opened a carbonated beverage. Nobody slowed down the camera.

The Consequence

Five solute systems. Water. Legumes. Herbs. Egg whites. And now a carbonated beverage at room temperature with zero thermal gradient. The SVE framework does not require heat to manifest. Structured vesicular entities form spontaneously in CO_2-saturated liquid at ambient conditions and exhibit the full behavioral repertoire: cohesion, motility, buoyancy defiance, and interface-crossing integrity. The "bubbles" in your Pepsi are alive with structure. They are not gas. They are SVEs—and they behave like dolphins in a black sea.

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1 Abstract

Experiment 14 documents a population of entities in freshly poured chilled Pepsi Black that exhibit behaviors structurally impossible for free CO_2 gas bubbles: (1) surface tension breach without rupture, (2) ballistic parabolic arcs through the air phase, (3) re-entry into the liquid with structural integrity preserved, and (4) post-re-entry sinking in a liquid approximately 577 times denser than CO_2 gas. A single SVE is tracked across 40 consecutive slow-motion frames (5 collages $\times$ 8 frames) documenting a complete "dolphin cycle": ascent $\rightarrow$ surface breach $\rightarrow$ airborne arc $\rightarrow$ apex $\rightarrow$ descent $\rightarrow$ re-entry dive $\rightarrow$ deep submersion $\rightarrow$ disappearance. No thermal gradient is applied. This extends the SVE framework to carbonated beverages at ambient conditions and demonstrates that structured vesicular entities form spontaneously in CO_2-saturated systems without requiring heat input.

2 Experimental Protocol

Parameter	Value
Container	Standard glass cup (copo de vidro)
Liquid	Pepsi Black, chilled, freshly served
Heat source	None — ambient conditions only
Thermal gradient	None applied
Recording	iPhone slow-motion video
Analysis	Frame-by-frame tracking of single SVE across 40 consecutive frames
Annotation	Red circle marking SVE position in each frame

Video: Supplementary Video — Experiment 14

3 Observational Data (5 Collages, 40 Frames)

Coll.	Fr	Frame #	Time	Phase	Observation
C1	F1	226	07:915	CONTEXT	SVE submerged deep inside the liquid
C1	F2	433	14:400	SUBMERGED	SVE ascending toward the surface
C1	F3	434	14:433	SUBMERGED	Continued upward approach
C1	F4	435	14:467	INTERFACE	SVE reaches liquid-air interface
C1	F5	436	14:500	EMERGING	Breaches surface — intact, no rupture
C1	F6	437	14:533	EMERGING	Above liquid level, spherical form preserved
C1	F7	438	14:567	AIRBORNE	SVE airborne, rising
C1	F8	439	14:600	AIRBORNE	Ascending arc trajectory
C2	F9	440	14:633	AIRBORNE	Significant height above surface
C2	F10	441	14:667	AIRBORNE	Approaching apex
C2	F11	442	14:700	APEX	Maximum height — peak of dolphin arc
C2	F12	443	14:733	DESCENDING	Arc curving downward
C2	F13	444	14:767	DESCENDING	Descent toward liquid
C2	F14	445	14:833	DESCENDING	Approaching liquid surface
C2	F15	446	14:867	DESCENDING	Nearing interface
C2	F16	447	14:900	DESCENDING	Approaching re-entry
C3	F17	448	14:933	DESCENDING	Final descent
C3	F18	449	14:967	DESCENDING	Near liquid surface
C3	F19	450	15:000	DESCENDING	Imminent re-entry
C3	F20	451	15:033	INTERFACE	Contact with liquid from above
C3	F21	452	15:067	RE-ENTRY	Penetrates into liquid — the “dive”
C3	F22	453	15:100	RE-ENTRY	Submerging, integrity maintained
C3	F23	454	15:133	SUBMERGED	Below the surface
C3	F24	455	15:167	SUBMERGED	Descending deeper
C4	F25	456	15:200	SUBMERGED	Sinking inside the liquid
C4	F26	457	15:233	SUBMERGED	Continued descent
C4	F27	458	15:267	SUBMERGED	Deeper — not returning to surface
C4	F28	459	15:300	SINKING	Below initial ascent point
C4	F29	460	15:333	SINKING	Deep inside liquid
C4	F30	461	15:367	SINKING	Gravitational behavior, not buoyant
C4	F31	462	15:400	SINKING	Near bottom of visible region
C4	F32	463	15:433	SINKING	At depth
C5	F33	464	15:467	DEEP	Fading from visibility
C5	F34	465	15:500	DEEP	Diminishing contrast
C5	F35	466	15:533	DEEP	Barely distinguishable
C5	F36	467	15:567	DEEP	Maximum tracked depth
C5	F37	469	15:600	FADING	Nearly lost
C5	F38	470	15:633	FADING	At tracking limit
C5	F39	471	15:667	FADING	Disappearing
C5	F40	472	15:700	GONE	SVE no longer trackable

4 Anomaly Analysis

4.1 Anomaly E1: Surface Tension Breach Without Rupture

At frames 435–436, the SVE crosses the liquid-air interface from below. For a gas bubble, crossing this interface means the thin liquid film is stretched beyond its mechanical limit and ruptures—this is the defining event in the bubble lifecycle. The Weber number criterion for bubble rupture at the free surface is:

$$\mathrm{We} = \frac{\rho_l v^2 d}{\sigma} > \mathrm{We}_{\mathrm{crit}} \implies \text{film rupture} \tag{1}$$

where ρ_l is liquid density, v is approach velocity, d is bubble diameter, and σ is surface tension. The classical prediction is binary: bubbles that reach the surface with sufficient energy breach the film and pop; those that don't, remain trapped below. There is no classical mechanism for a gas bubble to cross the interface intact while maintaining its spherical geometry. The observed behavior requires mechanical cohesion independent of the liquid film—consistent only with a solid or semi-solid structure (SVE).

4.2 Anomaly E2: Ballistic Parabolic Arc

Across frames 436–455, the SVE executes a complete parabolic trajectory through the air phase:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{g}\,t^2 \tag{2}$$

This ballistic arc requires three simultaneous properties: (1) structural integrity to survive surface breach, (2) sufficient inertial mass to follow a gravitational parabola, and (3) sufficient cohesion to survive atmospheric transit and re-entry impact. For a CO_2 gas pocket at atmospheric pressure, none of these properties exist. The gas would disperse into the atmosphere upon film rupture. The observed arc spans approximately 20 frames ($\sim$0.67 s at 30 fps playback of slow-motion footage), reaching a height visibly greater than the entity's own diameter—a macroscopic ballistic trajectory.

4.3 Anomaly E3: Re-Entry Without Dissolution

At frames 451–453, the SVE impacts the liquid surface from above and penetrates back into the liquid. Classical fluid dynamics predicts two outcomes for a gas entity contacting a liquid surface from above:

(a) It sits on the surface, supported by surface tension (floating);

(b) It merges with the gas phase above and the liquid closes.

There is no classical mechanism for a gas entity to dive *into* a liquid from above while maintaining structural identity. The observed "dive" is mechanically identical to a solid object entering water—requiring density, cohesion, and momentum transfer through the interface. This is the third interface crossing for the tracked SVE (liquid→air at breach, air→liquid at re-entry), each one independently falsifying the gas-bubble model.

4.4 Anomaly E4: Post-Dive Sinking

After re-entry (frames 456–472), the SVE does not return to the surface. It sinks progressively deeper until it disappears. The buoyancy force on a CO_2 gas bubble is:

$$F_b = (\rho_l - \rho_g)Vg \approx \rho_l V g \quad \text{since } \rho_g \approx 1.8 \text{ kg/m}^3 \ll \rho_l \approx 1040 \text{ kg/m}^3 \tag{3}$$

The buoyancy-to-weight ratio is:

$$\frac{F_b}{F_g} = \frac{\rho_l}{\rho_g} \approx \frac{1040}{1.8} \approx 577 \tag{4}$$

A CO_2 gas bubble experiences an upward force 577 times its own weight. Sinking is physically impossible. The observed sinking requires $\rho_{\text{SVE}} > \rho_l \approx 1040$ kg/m^3—the entity must be denser than the surrounding liquid. This is consistent with a structured vesicular entity containing liquid/solid phases, not free gas.

4.5 Anomaly E5: Population-Level Solid-Body Behavior

The tracked SVE is representative of a population. The full video documents dozens of entities simultaneously exhibiting:

- **Surface walking:** entities translating horizontally across the liquid-air interface without sinking or popping;
- **Bouncing:** entities impacting the surface and rebounding ("skipping stones on a lake");
- **Jumping:** entities breaching the surface and becoming briefly airborne;
- **Diving:** entities re-entering the liquid after airborne excursions.

This behavioral repertoire—walking, bouncing, jumping, diving—constitutes a population-level demonstration of solid-body mechanics. No gas bubble exhibits any of these behaviors. The phenomenon is not stochastic; it is systematic and reproducible in any freshly opened carbonated beverage.

5 Systematic Falsification

Classical Prediction	Observation	Status
Gas bubbles rupture upon breaching the liquid-air interface	SVE crosses interface intact, maintains spherical form, becomes airborne	FALSIFIED
Upon rupture, gas disperses irreversibly into the atmosphere	SVE executes complete ballistic arc and returns to the liquid	FALSIFIED
Gas entity contacting liquid from above either floats or merges	SVE dives into the liquid maintaining structural integrity	FALSIFIED
CO_2 gas ($\rho \approx 1.8$ kg/m^3) in liquid ($\rho \approx 1040$ kg/m^3): bubbles always float	SVE sinks progressively deeper after re-entry	FALSIFIED
Carbonated fizz = free CO_2 gas bubbles	Population of entities exhibiting solid-body mechanics	FALSIFIED

6 Discussion

6.1 No Thermal Gradient Required

Experiment 14 is the first in the Kitchen Thermodynamics series conducted at ambient temperature with zero applied thermal gradient. All previous experiments (1–13) involved active heating or cooling. The SVE behaviors observed here—structural cohesion, ballistic trajectories, buoyancy defiance—occur spontaneously in a carbonated liquid at refrigerator temperature. This demonstrates that the SVE framework is not limited to thermally driven systems; structured vesicular entities form wherever dissolved gas comes out of solution under appropriate conditions.

6.2 The CO_2 "Bubble" Is Not a Bubble

The classical model of carbonation is simple: CO_2 dissolved under pressure nucleates as gas bubbles when pressure drops, and these bubbles rise and pop. Experiment 14 falsifies every step of this narrative after nucleation. The entities that form are not gas pockets—they are

structured, mechanically cohesive objects with effective densities exceeding that of the surrounding liquid, capable of surviving repeated liquid-air-liquid interface crossings, and exhibiting ballistic trajectories consistent with solid-body dynamics. The term "bubble" is a misnomer. These are SVEs.

6.3 Cross-System Universality

Experiment 14 extends the SVE framework to a seventh distinct system:

System	Experiment	Anomaly Domain	Key Finding
Butter	Exp. 1	Motion	Climbs inclined planes against gravity
Water/Ice	Exp. 2	Phase	Sustained triple-point coexistence
Water/Milk	Exp. 3–4	Motion	Motion tracks flame, not T
Legumes	Exp. 5–6	Temperature	T drops during heating
Herbs	Exp. 11	Vapor + T	Vapor at 29.5°C; 23× post-flame rate
Egg whites	Exp. 13	Vapor + nucleation	Vapor suppressed at 100°C
Pepsi Black	**Exp. 14**	**Mechanics**	**SVEs jump, dive, sink — no thermal gradient**

The universality is now established across thermal and non-thermal systems, organic and inorganic substrates, heated and ambient conditions. SVEs are not a thermal curiosity—they are a fundamental feature of liquid systems containing dissolved or entrapped gas.

6.4 Accessibility and Reproducibility

Experiment 14 is the most reproducible and accessible in the series. Required materials: one bottle of any carbonated beverage (Pepsi Black, Coca-Cola, sparkling water—any), one glass, one smartphone with slow-motion capability. No stove, no thermometer, no laboratory equipment. The phenomenon is visible in every glass of soda ever poured. The barrier to independent replication is essentially zero.

7 Conclusion

Experiment 14 documents structured vesicular entities in chilled Pepsi Black that breach the liquid-air interface without rupturing, execute ballistic parabolic arcs through the air, re-enter the liquid maintaining structural integrity, and sink in a liquid 577 times denser than CO_2 gas. A single SVE tracked across 40 consecutive frames (5 collages) completes a full "dolphin cycle"—the most visually dramatic demonstration of SVE structural integrity in the Kitchen Thermodynamics series. No thermal gradient is applied. The "CO_2 bubbles" in carbonated beverages are not gas bubbles. They are structured vesicular entities—and they behave like dolphins in a black sea.

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Supplementary Materials

Video: youtu.be/FBsfnDhfJIk

All supplementary materials and complete experimental archive available at: Free Science: Google Drive